SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE

Project Title: HUFT – E-Commerce Order Management Database System

1.BASIC SQL QUERIES

1.1 SELECT query

The SELECT statement is used to retrieve data from a table.

1.1.1 Display all products

```
SQL> SELECT *
  2  FROM HuftProduct;

PRODUCT_ID PRODUCT_NAME                  CATEGORY_ID      PRICE      STOCK
---------- ------------------------------ ----------- ---------- ----------
       101 Chicken Dog Food                         1     599.00         50
       102 Grain Free Cat Food                      2     749.00         35
       103 Leather Dog Collar                       3     450.00         25
       104 Pet Grooming Shampoo                     4     350.00         40
       106 Cat Litter                               2     550.00         30
       107 Dental Chew Sticks                       1     399.00         45
       108 Pet Feeding Bowl                         3     275.00         20
       109 Puppy Training Pads                      1     499.00         55
       110 Pet Conditioner                          4     425.00         30
       112 Cat Grooming Brush                       4     325.00         28
       113 Dog Harness                              3     699.00         22
       114 Premium Dog Treats                       1     450.00         40

12 rows selected.
```

Task 1.1.2 – Display Selected Product Details

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM HuftProduct;

PRODUCT_ID PRODUCT_NAME                      PRICE
---------- ------------------------------ ----------
       101 Chicken Dog Food                599.00
       102 Grain Free Cat Food             749.00
       103 Leather Dog Collar              450.00
       104 Pet Grooming Shampoo            350.00
       106 Cat Litter                      550.00
       107 Dental Chew Sticks              399.00
       108 Pet Feeding Bowl                275.00
       109 Puppy Training Pads             499.00
       110 Pet Conditioner                 425.00
       112 Cat Grooming Brush              325.00
       113 Dog Harness                     699.00
       114 Premium Dog Treats              450.00

12 rows selected.
```

Task 1.1.3 – Find Products with Price Less Than ₹1000

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM HuftProduct
  3  WHERE Price < 1000;

PRODUCT_ID PRODUCT_NAME                    PRICE
---------- ------------------------------ -----------
       101 Chicken Dog Food                599.00
       102 Grain Free Cat Food             749.00
       103 Leather Dog Collar              450.00
       104 Pet Grooming Shampoo            350.00
       106 Cat Litter                      550.00
       107 Dental Chew Sticks              399.00
       108 Pet Feeding Bowl                275.00
       109 Puppy Training Pads             499.00
       110 Pet Conditioner                 425.00
       112 Cat Grooming Brush              325.00
       113 Dog Harness                     699.00
       114 Premium Dog Treats              450.00

12 rows selected.
```

Task 1.1.4 – Find Products with Price Greater Than ₹100

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM HuftProduct
  3  WHERE Price > 100;

PRODUCT_ID PRODUCT_NAME                    PRICE
---------- ------------------------------ -----------
       101 Chicken Dog Food                599.00
       102 Grain Free Cat Food             749.00
       103 Leather Dog Collar              450.00
       104 Pet Grooming Shampoo            350.00
       106 Cat Litter                      550.00
       107 Dental Chew Sticks              399.00
       108 Pet Feeding Bowl                275.00
       109 Puppy Training Pads             499.00
       110 Pet Conditioner                 425.00
       112 Cat Grooming Brush              325.00
       113 Dog Harness                     699.00
       114 Premium Dog Treats              450.00

12 rows selected.
```

Task 1.1.5 – Products from Lowest to Highest Price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM HuftProduct
  3  ORDER BY Price ASC;

PRODUCT_ID PRODUCT_NAME                         PRICE
---------- ------------------------------ ----------
       108 Pet Feeding Bowl                      275.00
       112 Cat Grooming Brush                    325.00
       104 Pet Grooming Shampoo                  350.00
       107 Dental Chew Sticks                    399.00
       110 Pet Conditioner                       425.00
       114 Premium Dog Treats                    450.00
       103 Leather Dog Collar                    450.00
       109 Puppy Training Pads                   499.00
       106 Cat Litter                            550.00
       101 Chicken Dog Food                      599.00
       113 Dog Harness                           699.00
       102 Grain Free Cat Food                   749.00

12 rows selected.
```

Task 1.1.6 – Products from Highest to Lowest Price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM HuftProduct
  3  ORDER BY Price DESC;

PRODUCT_ID PRODUCT_NAME                         PRICE
---------- ------------------------------ ----------
       102 Grain Free Cat Food                   749.00
       113 Dog Harness                           699.00
       101 Chicken Dog Food                      599.00
       106 Cat Litter                            550.00
       109 Puppy Training Pads                   499.00
       103 Leather Dog Collar                    450.00
       114 Premium Dog Treats                    450.00
       110 Pet Conditioner                       425.00
       107 Dental Chew Sticks                    399.00
       104 Pet Grooming Shampoo                  350.00
       112 Cat Grooming Brush                    325.00
       108 Pet Feeding Bowl                      275.00

12 rows selected.
```

Task 1.1.7 – Display Unique Product Categories

```
SQL> SELECT DISTINCT
  2      c.Category_ID,
  3      c.Category_Name
  4  FROM HuftProduct p
  5  JOIN HuftCategory c
  6      ON p.Category_ID = c.Category_ID
  7  ORDER BY c.Category_ID;

CATEGORY_ID CATEGORY_NAME
----------- --------------------------------------------------
          1 Dog Food
          2 Cat Food
          3 Accessories
          4 Grooming
```

Task 2.1 – Search Products Based on Price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM HuftProduct
  3  WHERE Price BETWEEN 100 AND 400
  4  ORDER BY Price ASC;

PRODUCT_ID PRODUCT_NAME                    PRICE
---------- ------------------------------ -----------
       108 Pet Feeding Bowl                275.00
       112 Cat Grooming Brush              325.00
       104 Pet Grooming Shampoo            350.00
       107 Dental Chew Sticks              399.00
```

Task 2.2 – Search Products Based on Category

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      c.Category_Name,
  5      p.Price
  6  FROM HuftProduct p
  7  JOIN HuftCategory c
  8      ON p.Category_ID = c.Category_ID
  9  WHERE c.Category_ID = 1;

PRODUCT_ID PRODUCT_NAME                   CATEGORY_NAME                                      PRICE
---------- ------------------------------ -------------------------------------------------- -----------
       101 Chicken Dog Food               Dog Food                                           599.00
       107 Dental Chew Sticks             Dog Food                                           399.00
       109 Puppy Training Pads            Dog Food                                           499.00
       114 Premium Dog Treats             Dog Food                                           450.00
```

Task 3.1 – Display All Customers

```
SQL> SELECT *
  2  FROM CUST3;

       CID CNAME                          EMAIL                          PHONE_NO ADDRESS
---------- ------------------------------ ------------------------------ ---------- ------------------------------------
         1 Abi                            abi@gmail.com                  1234567890 Ramapuram Chennai
         2 Anu                            anu@gmail.com                   987654321 Madurai
         3 Ravi                           ravi@gmail.com                 9087654321 Chengalpattu
         4 Karthik                        karthik@gmail.com              9876543210 Coimbatore
         5 Priya                          priya@gmail.com                8765432109 Salem
         6 Divya                          divya@gmail.com                7654321098 Trichy
         7 Arun                           arun@gmail.com                 6543210987 Vellore
         8 Meena                          meena@gmail.com                9432109876 Tirunelveli
        10 Nisha                          nisha@gmail.com                9210987654 Vellore

9 rows selected.
```

Task 3.2 – Search for a Particular Customer

```
SQL> SELECT *
  2  FROM CUST3
  3  WHERE CID = 1;

       CID CNAME                          EMAIL                          PHONE_NO ADDRESS
---------- ------------------------------ ------------------------------ ---------- ------------------------------------
         1 Abi                            abi@gmail.com                  1234567890 Ramapuram Chennai
```

Task 4.1 – Apply AND Condition

```
SQL> SELECT Product_ID, Product_Name, Price, Stock
  2  FROM HuftProduct
  3  WHERE Price < 500
  4  AND Stock > 20;

PRODUCT_ID PRODUCT_NAME                         PRICE      STOCK
---------- ------------------------------ ---------- ----------
       103 Leather Dog Collar                 450.00         25
       104 Pet Grooming Shampoo               350.00         40
       107 Dental Chew Sticks                 399.00         45
       109 Puppy Training Pads                499.00         55
       110 Pet Conditioner                    425.00         30
       112 Cat Grooming Brush                 325.00         28
       114 Premium Dog Treats                 450.00         40

7 rows selected.
```

Task 4.2 – Apply OR Condition

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      c.Category_Name,
  5      p.Price
  6  FROM HuftProduct p
  7  JOIN HuftCategory c
  8      ON p.Category_ID = c.Category_ID
  9  WHERE p.Category_ID = 1
 10     OR p.Category_ID = 2;

PRODUCT_ID PRODUCT_NAME                 CATEGORY_NAME                                      PRICE
---------- ---------------------------- -------------------------------------------- ----------
       101 Chicken Dog Food             Dog Food                                         599.00
       102 Grain Free Cat Food          Cat Food                                         749.00
       106 Cat Litter                   Cat Food                                         550.00
       107 Dental Chew Sticks           Dog Food                                         399.00
       109 Puppy Training Pads          Dog Food                                         499.00
       114 Premium Dog Treats           Dog Food                                         450.00

6 rows selected.
```

Task 4.3 – Apply IN Operator

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      c.Category_Name,
  5      p.Price
  6  FROM HuftProduct p
  7  JOIN HuftCategory c
  8      ON p.Category_ID = c.Category_ID
  9  WHERE p.Category_ID IN (1, 2);

PRODUCT_ID PRODUCT_NAME                 CATEGORY_NAME                                      PRICE
---------- ---------------------------- -------------------------------------------- ----------
       101 Chicken Dog Food             Dog Food                                         599.00
       102 Grain Free Cat Food          Cat Food                                         749.00
       106 Cat Litter                   Cat Food                                         550.00
       107 Dental Chew Sticks           Dog Food                                         399.00
       109 Puppy Training Pads          Dog Food                                         499.00
       114 Premium Dog Treats           Dog Food                                         450.00

6 rows selected.
```

Task 4.4 – Apply BETWEEN Operator

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM HuftProduct
  3  WHERE Price BETWEEN 100 AND 400
  4  ORDER BY Price ASC;

PRODUCT_ID PRODUCT_NAME                      PRICE
---------- ------------------------------ ----------
       108 Pet Feeding Bowl                  275.00
       112 Cat Grooming Brush                325.00
       104 Pet Grooming Shampoo              350.00
       107 Dental Chew Sticks                399.00
```

Task 5.1 – Product Price Report

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM HuftProduct
  3  ORDER BY Price DESC;

PRODUCT_ID PRODUCT_NAME                        PRICE
---------- ------------------------------ ----------
       102 Grain Free Cat Food                749.00
       113 Dog Harness                        699.00
       101 Chicken Dog Food                   599.00
       106 Cat Litter                         550.00
       109 Puppy Training Pads                499.00
       103 Leather Dog Collar                 450.00
       114 Premium Dog Treats                 450.00
       110 Pet Conditioner                    425.00
       107 Dental Chew Sticks                 399.00
       104 Pet Grooming Shampoo               350.00
       112 Cat Grooming Brush                 325.00
       108 Pet Feeding Bowl                   275.00

12 rows selected.
```

Task 5.2 – Customer Order Report

```
SQL> SELECT
  2      c.CID AS Customer_ID,
  3      c.CNAME AS Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      o.Total_Amount,
  7      o.Order_Status
  8  FROM CUST3 c
  9  JOIN HuftOrder o
 10      ON c.CID = o.Customer_ID
 11  ORDER BY c.CID, o.Order_Date;

CUSTOMER_ID CUSTOMER_NAME                  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
----------- ------------------------------ ---------- --------- ------------ --------------------
          1 Abi                                  1001 18-SEP-26         1750 Delivered
          2 Anu                                  1002 18-SEP-26         1849 Shipped
          3 Ravi                                 1003 19-SEP-26         1099 Processing
          4 Karthik                              1004 19-SEP-26         1273 Delivered
          5 Priya                                1005 20-SEP-26         1075 Shipped
          6 Divya                                1006 20-SEP-26          650 Processing
          7 Arun                                 1007 20-SEP-26         1197 Delivered
          8 Meena                                1008 21-SEP-26         1498 Processing
         10 Nisha                                1010 21-SEP-26          900 Processing

9 rows selected.
```

Task 5.3 – Product Order Report

```
SQL> SELECT
  2      p.Product_ID,
  3      p.Product_Name,
  4      o.Order_ID,
  5      oi.Quantity,
  6      oi.Price,
  7      oi.Total_Amount
  8  FROM HuftProduct p
  9  JOIN HuftOrder_Item oi
 10      ON p.Product_ID = oi.Product_ID
 11  JOIN HuftOrder o
 12      ON oi.Order_ID = o.Order_ID
 13  ORDER BY p.Product_ID, o.Order_ID;
```

PRODUCT_ID	PRODUCT_NAME	ORDER_ID	QUANTITY	PRICE	TOTAL_AMOUNT
101	Chicken Dog Food	1001	2	650.00	1300
101	Chicken Dog Food	1006	1	650.00	650
102	Grain Free Cat Food	1002	1	749.00	749
102	Grain Free Cat Food	1008	2	749.00	1498
103	Leather Dog Collar	1001	1	450.00	450
103	Leather Dog Collar	1010	2	450.00	900
104	Pet Grooming Shampoo	1003	2	350.00	700
106	Cat Litter	1002	2	550.00	1100
107	Dental Chew Sticks	1003	1	399.00	399
107	Dental Chew Sticks	1007	3	399.00	1197
108	Pet Feeding Bowl	1004	2	275.00	550
109	Puppy Training Pads	1004	1	499.00	499
109	Puppy Training Pads	1009	2	499.00	998
110	Pet Conditioner	1005	1	425.00	425
112	Cat Grooming Brush	1005	2	325.00	650

15 rows selected.

Task 5.4 – Customer-wise Total Orders

```
SQL> SELECT
  2      c.CID AS Customer_ID,
  3      c.CNAME AS Customer_Name,
  4      COUNT(o.Order_ID) AS Total_Orders
  5  FROM CUST3 c
  6  LEFT JOIN HuftOrder o
  7      ON c.CID = o.Customer_ID
  8  GROUP BY c.CID, c.CNAME
  9  ORDER BY c.CID;

CUSTOMER_ID CUSTOMER_NAME                  TOTAL_ORDERS
----------- ------------------------------ ------------
          1 Abi                                       1
          2 Anu                                       1
          3 Ravi                                      1
          4 Karthik                                   1
          5 Priya                                     1
          6 Divya                                     1
          7 Arun                                      1
          8 Meena                                     1
         10 Nisha                                     1

9 rows selected.
```

Task 5.5 – Customer-wise Total Purchase Amount

```
SQL> SELECT
  2      c.CID AS Customer_ID,
  3      c.CNAME AS Customer_Name,
  4      SUM(o.Total_Amount) AS Total_Purchase_Amount
  5  FROM CUST3 c
  6  JOIN HuftOrder o
  7      ON c.CID = o.Customer_ID
  8  GROUP BY
  9      c.CID,
 10      c.CNAME
 11  ORDER BY c.CID;

CUSTOMER_ID CUSTOMER_NAME                  TOTAL_PURCHASE_AMOUNT
----------- ------------------------------ ---------------------
          1 Abi                                             1750
          2 Anu                                             1849
          3 Ravi                                            1099
          4 Karthik                                         1273
          5 Priya                                           1075
          6 Divya                                            650
          7 Arun                                            1197
          8 Meena                                           1498
         10 Nisha                                            900

9 rows selected.
```